

# Global and Country-Specific Output Growth Uncertainty and Macroeconomic Performance\*

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## Abstract

We identify global and country-specific measures of output growth uncertainty for a large OECD country sample by means of a dynamic factor model with stochastic volatility. We find evidence for major bouts of global uncertainty in the early 1970s and late 2000s, and a number of periods with elevated levels of either global or national uncertainty, particularly in the early 1980s, 1990s and 2000s. VAR impulse responses of national macroeconomic variables to our estimated measures of uncertainty reveal that global uncertainty is the major driver of macroeconomic performance in most countries, whereas the impact of national uncertainty is small and frequently insignificant. We also find that uncertainty is transmitted primarily through investment and trade flows rather than through consumption demand.

## I. Introduction

The notion of economic uncertainty as a source of macroeconomic fluctuations is associated most prominently with the writings of John Maynard Keynes. Renewed interest in the potential links between uncertainty and macroeconomic performance and their quantitative importance has spawned a recent and blossoming literature based on rigorous theoretic and empirical modeling. However, so far this literature mostly focuses on the national rather than the global dimension of uncertainty. In this paper, we identify global and country-specific measures of output growth uncertainty for a large OECD country sample by means of a dynamic factor model (DFM) with stochastic volatility. We find evidence for major bouts of global uncertainty in the early 1970s and late 2000s, and a number of periods with elevated levels of either global or national uncertainty, particularly in the early 1980s, 1990s and 2000s. VAR impulse responses of national macroeconomic variables to our estimated

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measures of uncertainty reveal that global uncertainty is the major driver of macroeconomic performance in most countries, whereas the impact of national uncertainty is small and frequently insignificant. We also find that uncertainty is transmitted primarily through investment and trade flows rather than through consumption demand in most sample countries.

A growing literature investigates the potential links between macroeconomic uncertainty and economic performance. One strand of that literature uses GARCH models to proxy macroeconomic uncertainty by the conditional variance of unanticipated shocks to the time series of inflation and output growth (e.g. Grier and Perry, 1996; Shields *et al.*, 2005; Fountas and Karanasos, 2007). These papers detect one-way or two-way causal relationships between many of the possible bilateral combinations of inflation, output growth and their respective second moments. In particular, increased growth uncertainty is frequently found to cause significantly lower average growth rates (Grier *et al.*, 2004; Bredin and Fountas, 2009).

A different strand of recent literature uses firm-level data with a focus on analysing the impact of uncertainty shocks on the real economy (see e.g. Bloom, 2009; Arellano, Bai and Kehoe, 2012; Bloom *et al.*, 2012; Christiano, Motto and Rostagno, 2014). Following up on the theory of investment under uncertainty (e.g. Dixit and Pindyck, 1994), this literature identifies real-options effects as an important transmission channel of uncertainty on the real economy.<sup>1</sup> The real-options effect is associated with irreversibility in investment, implying that heightened uncertainty makes firms more cautious about hiring and investing, and consumers more cautious about buying durables. In an influential study using US data Bloom (2009) finds that macroeconomic uncertainty shocks generate recessions because a higher uncertainty causes firms to temporarily suspend investment and hiring. Carrière-Swallow and Céspedes (2013) report similar evidence of adverse effects of uncertainty on investment and consumption for a large sample consisting of both developed and developing countries.

A number of studies provide theoretical arguments and empirical evidence for an analogous uncertainty effect on international trade flows. Handley (2014) and Handley and Limão (2012) show that with sunk costs of entry, trade policy uncertainty raises the real-options value of waiting to enter export markets. Novy and Taylor (2014) provide an explanation for the global trade collapse of 2008–09 by considering a model in which fixed costs of ordering let firms hold an inventory of imported intermediates. In their framework, uncertainty shocks induce firms to cut their orders of foreign intermediates disproportionately strongly, thereby inducing bigger contractions in international trade flows compared to domestic economic activity.

A common feature of the literature cited above is their sole focus on the national incidence of uncertainty in individual countries. So far, almost no attempt has been made to differentiate between its national and global dimensions.<sup>2</sup> Yet global uncertainty may

<sup>1</sup> Some studies also allow for reverse causation, in which uncertainty is endogenously influenced by first-moment business cycle shocks (e.g. Van Nieuwerburgh and Veldkamp, 2006; Bachmann and Moscarini, 2012; D'Erasmus and Moscoso-Boedo, 2013).

<sup>2</sup> To the best of our knowledge, there are only two exceptions. Nakamura, Sergeyev and Steinsson (2012) identify a world stochastic volatility process using over 100 years of consumption data across a large sample of countries. However, their focus is not so much on macroeconomic performance, but rather on explaining a number of asset pricing anomalies. The other paper being that of Gourio, Siemer and Verdelhan (2013) who analyse various aspects

constitute an independent and important determinant of the international reach of financial and economic crises. Potential drivers of global uncertainty can be thought of as including all factors with a simultaneous impact on national levels of uncertainty, like oil, commodity and asset price shocks, concurrent changes in national monetary or fiscal policies, or synchronous shifts of investor sentiment possibly associated with changes in global perceptions about financial, economic or political risk.<sup>3</sup>

In order to identify global macroeconomic uncertainty, we draw on the literature of international co-movements of macroeconomic variables using DFMs. These models have been utilized to detect international business cycles (Kose, Prasad and Terrones, 2003; Crucini, Kose and Otrok, 2011; Kose, Otrok and Prasad, 2012) or global inflation dynamics (Ciccarelli and Mojon, 2010; Mumtaz and Surico, 2012). In this literature, co-movement is defined as common shocks to the mean. No attempt has yet been made to analyse global macroeconomic uncertainty in terms of the second moments of common shocks.

Time-varying uncertainty shocks are unobservable, so a range of proxies is used in empirical work. These include measures of stock market volatility (e.g. Bloom, 2009), the cross-sectional dispersion of plant-level shocks to total factor productivity (e.g. Bloom *et al.*, 2012), survey-based forecaster disagreements (e.g. Bachmann, Elstner and Sims, 2013), or the appearance of uncertainty-related keywords in news publications (e.g. Alexopoulos and Cohen, 2009; Baker, Bloom and Davis, 2013). An alternative is to apply time series models to obtain uncertainty proxies. The two approaches most frequently utilized are stochastic volatility and *GARCH* models. Although there appears to be a proliferation of the latter, both approaches have been shown to yield comparable results relative to survey-based measures of inflation and output growth uncertainty (see Chua, Kim and Suardi, 2011; Grimme, Henzel and Wieland, 2011). However, whereas the volatility process in *GARCH* models is explained solely in terms of level changes, stochastic volatility models are more flexible as they allow for a separate innovation impinging on volatility, independent of any changes in levels (Fernández-Villaverde and Rubio-Ramírez, 2010).

This paper is concerned with the empirical identification of global macroeconomic uncertainty. Our approach consists of setting up a DFM to decompose time series on output growth for a large OECD country sample into common and country-specific components. We analyse VAR impulse response functions to compare the effects of global and national uncertainty shocks on national economic performance. We also investigate the potential role of investment, consumption and trade as transmission channels of uncertainty shocks. The remainder of the paper is structured as follows: section II presents the empirical model and estimation strategy to identify global and national output growth uncertainty. Section III reports our empirical uncertainty estimates and relates the results to the occurrence of a number of major uncertainty events. Section IV presents the results of the VAR impulse response analysis. A final section concludes.

related to international risk cycles, and also whether global uncertainty has a significant effect on macroeconomic aggregates, but their measure is exclusively based on global equity market volatility across G7 countries.

<sup>3</sup> An example of such synchronous shifts in investor sentiment has recently been provided by Bacchetta, Tille and van Wincoop (2012) who develop the concept of self-fulfilling risk panics. A weak fundamental like the health of financial institutions in the U.S. or the scale of debt in Greece suddenly becomes a focal point of fear everywhere. This fundamental then takes on the role of a coordination device for a self-fulfilling shift in risk perceptions.

## II. Empirical model and estimation strategy

### A dynamic latent factor model

We set up a latent DFM to decompose output growth in each country into a common (or global) and country-specific factors. The conditional factor variances are modeled as stochastic volatility processes and interpreted as reflecting uncertainty in the underlying factor.

More specifically, the DFM is given by

$$y_{it} = \beta_i^0 + \beta_i^1 \pi_{it} + \Gamma_i R_t + I_{it} + \epsilon_{it} \quad (1)$$

where  $y_{it}$  denotes output growth in country  $i$  at time  $t$ , with  $i = 1, \dots, N$  and  $t = 1, \dots, T$ , with  $\beta_i^0$  a country-specific constant. The global factor  $R_t$  reflects common fluctuations in output growth across all sample countries, and  $\Gamma_i$  denotes the country-specific factor loadings that capture the extent to which output growth in country  $i$  moves with the global factor.  $I_{it}$  is the country-specific idiosyncratic component of nation  $i$ 's output growth rate, and  $\pi_{it}$  is included as an exogenous regressor to control for price effects. Its impact is captured by the country-specific coefficient  $\beta_i^1$ . The country-specific error term,  $\epsilon_{it}$ , accounts for measurement error. It is assumed to be  $\epsilon_{it} \sim iid\mathcal{N}(0, \sigma_{\epsilon_{it}})$ . Neither the scales nor the signs of the factors and factor loadings in our model are separately identified. To normalize the signs of the factor loadings and factors, we impose the factor loadings to be positive,  $\Gamma_i > 0$ . To solve the scale problem, we impose an unconditional variance of unity on the shocks of the common factor.

The dynamics of the model are captured by modeling the factors as stationary autoregressive processes. We assume the common factor to follow an independent AR( $p$ ) process<sup>4</sup>:

$$R_t = \sum_{k=1}^p \rho_k R_{t-k} + \exp(h_t) \xi_t, \quad \xi_t \sim iid\mathcal{N}(0, 1) \quad (2)$$

The idiosyncratic factors are also assumed to evolve according to an AR( $p$ ) process:

$$I_{it} = \sum_{k=1}^p \Phi_k^i I_{it-k} + \exp(g_{it}) \eta_{it}, \quad \eta_{it} \sim iid\mathcal{N}(0, \Sigma_i) \quad (3)$$

We assume the shocks in equations (2) and (3) to be uncorrelated, so that the common and country-specific factors are orthogonal. The volatilities of all factors are assumed to be time-varying and modeled to evolve as a stochastic volatility process following a random walk, i.e.

$$h_t = h_{t-1} + \gamma_t, \quad \gamma_t \sim iid\mathcal{N}(0, \sigma_\gamma^2) \quad (4)$$

$$g_{it} = g_{it-1} + \omega_{it}, \quad \omega_{it} \sim iid\mathcal{N}(0, \sigma_{\omega_i}^2) \quad (5)$$

<sup>4</sup>As is commonly done in the literature, we impose  $p = 2$  lags when estimating the model with quarterly data.

where  $h_t$  is the stochastic volatility component of the common factor and  $g_{it}$  the stochastic volatility pertaining to the country-specific factors. Equations (4) and (5) constitute our measure of global output growth uncertainty and country-specific uncertainty respectively.

### Estimation

Denote by  $R = \{R_t\}_{t=1}^T$  the common factor  $R_t$  stacked over the sample period. Similar notation is used for the idiosyncratic factors,  $I$ , the stochastic volatilities,  $h$  and  $g$ , and the observed data on output growth  $Y$ . We estimate the model using Bayesian Markov chain Monte Carlo (MCMC) methods to jointly sample the states  $(R, I, h, g)$  and the hyperparameters  $\Psi = \{\beta_i^0, \beta_i^1, \Gamma_i, \Phi_1^i, \dots, \Phi_p^i, \rho_1, \dots, \rho_p, \sigma_{\eta_i}^2, \sigma_\gamma^2, \sigma_{\omega_i}^2\}$ . More specifically, we use a Gibbs sampling algorithm, which is a simulation method for approximating the intractable joint and marginal posterior distributions of the unknown parameters and the unobserved states by sampling from conditional distributions. Thus, Gibbs sampling allows us to substantially simplify the estimation process by separating the states and parameters into several blocks that are tractable conditional on the other blocks.

The estimation process consists of several steps that we iterate repeatedly. We divide the model states and parameters into 4 blocks. The first block contains the common factor,  $R$ , and the country-specific factors,  $I$ , which we sample simultaneously conditional on the data, the stochastic volatilities,  $h$  and  $g$ , and the hyperparameters,  $\Psi$ . To sample  $R$  and  $I$  we use a forward-filtering and backward-sampling approach that is based on a general state space form of our model.<sup>5</sup> As the stochastic volatility components are mutually uncorrelated, we can separate the volatility series for the common component,  $h$ , and the idiosyncratic components,  $g$ , in two separate blocks. We follow Omori *et al.* (2007) and transform the nonlinear stochastic volatility components into linear components by taking the logarithm of their squares. Estimates of  $h$  and  $g$  are then obtained by using the forward-filtering and backward-sampling approach and apply it to the linearized model. Details on the estimation of the stochastic volatility components can be found in the Appendix. The final block contains the hyperparameters  $\Psi$ . These can be estimated and drawn given the common factor,  $R$ , and the country-specific factors,  $I$ , where the problem reduces to a standard static linear regression model.<sup>6</sup> We use inverse gamma priors for variance parameters and normal priors for all other parameters. All prior distributions are uninformative.

To summarize, we approximate the joint posterior density  $f(R, I, h, g, \Psi | Y)$  by iterating through the following steps  $J$  times:

1. assign arbitrary starting values  $(R^0, I^0, h^0, g^0, \Psi^0)$
2. sample  $R^1$  and  $I_i^1$  from  $f(R, I_i | Y_i, h^0, g_i^0, \Psi^0)$  for  $i = 1, \dots, N$
3. sample  $h^1$  from  $f(h | R^1, \Psi^0)$
4. sample  $g_i^1$  from  $f(g_i | Y, R^1, \Psi^0)$  for  $i = 1, \dots, N$
5. sample  $\Psi^1$  from  $f(\Psi | Y, R^1, h^1, g^1)$ .
6. go to step 2 and proceed using the most recent values of the conditioning parameters.

After a sufficiently long burn-in period  $B$ , the sequence of draws  $(B + 1, \dots, J)$  approximates a sample from the virtual posterior distribution.

<sup>5</sup> See Kim and Nelson (1999) for a textbook treatment of the Gibbs sampler for DFMs.

<sup>6</sup> For more details on this approach, see Bauwens, Lubrano and Richard (1999, pp. 56–61).

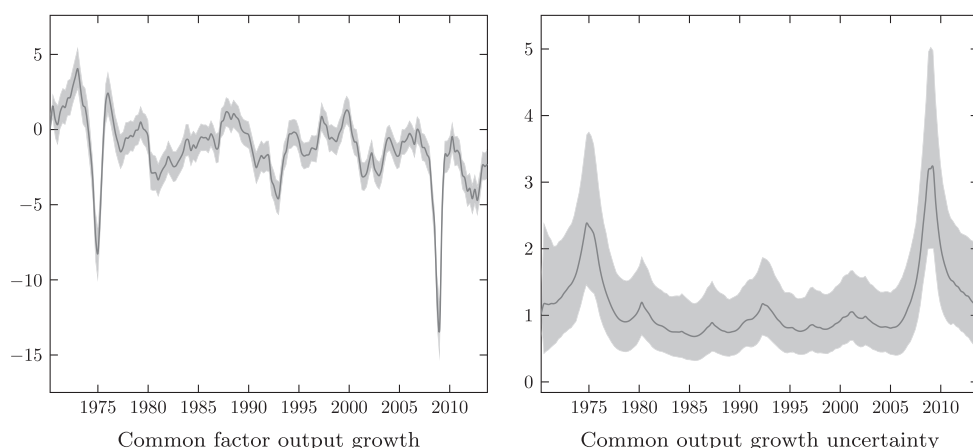


Figure 1. Global measures

Note: The shaded areas indicate 90% confidence intervals.

### III. Measures of uncertainty

We estimate our DFM using quarterly data on the annualized log change of GDP and CPI. Our sample includes 20 industrialized countries<sup>7</sup> and covers the period from 1970Q1 to 2013Q4. All data are obtained from the OECD Statistics database. The data are seasonally adjusted and demeaned.

Figure 1 visualizes the estimates of common output growth as well as the corresponding uncertainty measure. The global output growth factor displays two clearly visible global recessions. The first major slump occurs in the early 1970s around the time of the first oil crisis. The second and even more severe downturn is associated with the Great Recession following the US subprime crisis of 2007–09. The global uncertainty measure turns out to be strongly countercyclical, with a correlation coefficient between the global growth rate and the global uncertainty measure of  $-0.48$ . In particular, the two major global recessions are also associated with the strongest spikes of the global uncertainty measure. Apart from these two episodes, periods of heightened global uncertainty also occur during the early 1980s, 1990s and 2000s. The increase in global uncertainty in the early 1980s comes in the aftermath of the second spike in oil prices and may potentially also be due to uncertainties arising from a global shift towards more conservative monetary policy regimes (see e.g. Suda and Zervou, 2014). The 1990s increase in the global uncertainty measure coincides with the beginning of the Gulf War and the third oil price shock, whereas the increased uncertainty in the early 2000s comes in the wake of the bursting of the dotcom bubble, the global repercussions of the 9/11 terror attacks and the uncertainties surrounding the Enron scandal. Finally, there are also two smaller blips of heightened uncertainty in 1987 and 1998 following Black Monday and the Asian financial crisis respectively.

The estimated measures for country-specific uncertainty are shown in Figure 2. The graphs exhibit a substantial amount of heterogeneity across the sample countries. With the component common to all countries incorporated into the global uncertainty measure,

<sup>7</sup> Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, UK and the US.

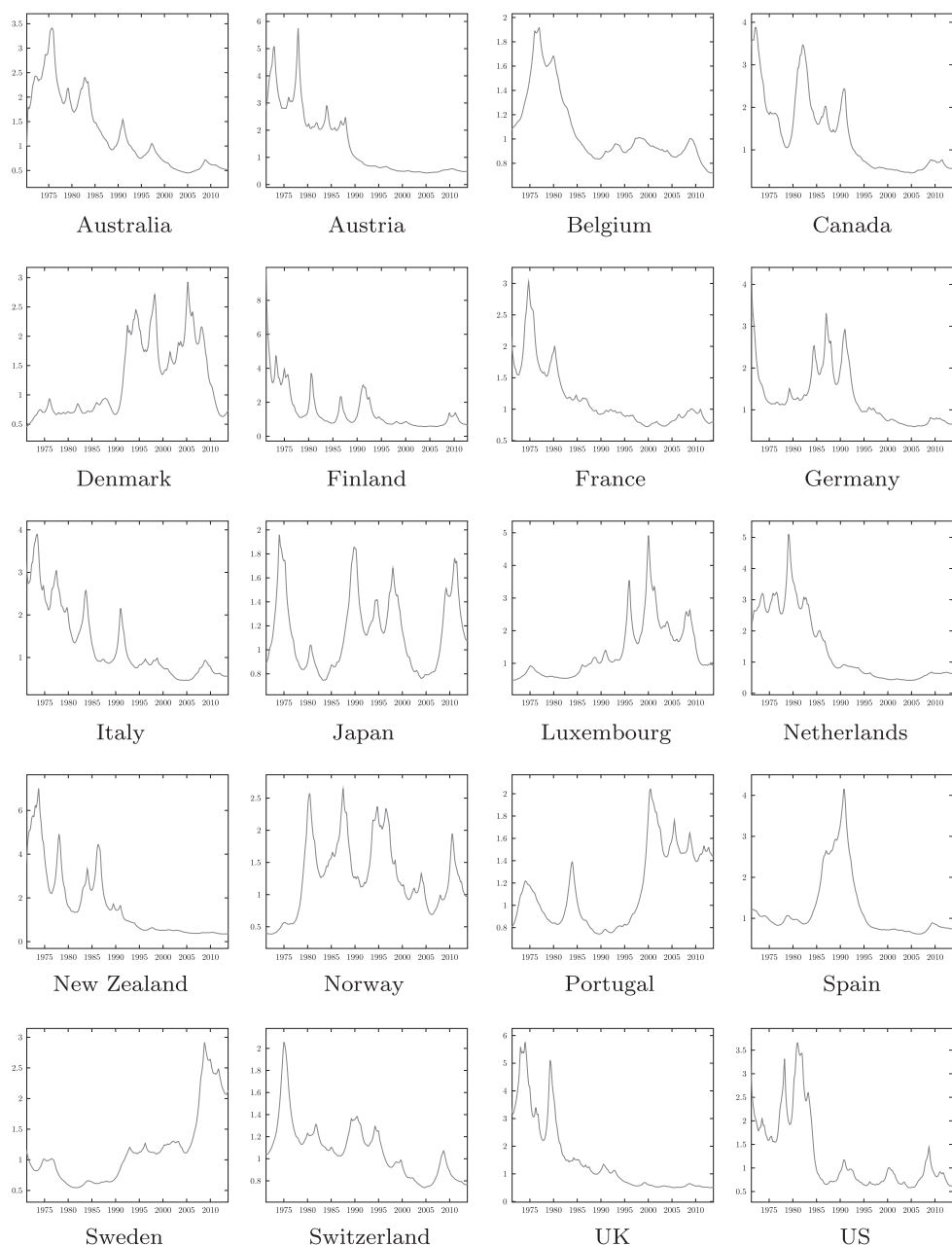


Figure 2. Country-specific output growth uncertainty

the national measures capture two kinds of uncertainty events. One comes in the form of uncertainty specific to individual countries, and the other represents the differential impact of global uncertainty shocks on individual countries. A comparison of the global and national incidence of uncertainty reveals that whereas the average level of global uncertainty is rather low during much of the 1980s, 1990s and early 2000s, a similar

pattern is found in the national uncertainty measures for some countries like the US and the UK, but is absent in many other countries.

Denmark, Luxembourg, Portugal and Sweden even display increases of national uncertainty in the latter part of the sample. These uncertainty patterns can be explained in part by recessions in some of these countries, like Sweden's in 2007, where economic growth slowed down right before the onset of the financial crisis, or Luxembourg's in 1992–96 and 2000–01, with the economic downturn of 2001 accentuated by weaknesses in the financial sector (Statec Luxembourg, 2003). In Portugal large imbalances built up in the second half of the 1990s, with a sharp rise in the net borrowing of the economy which became a major cause for concern (European Economy, 2004). The Portuguese economy also slipped into recession in the early 2000s, resulting in the weakest performance among EU member states. Finally, the surge in uncertainty in Denmark throughout the 1990s can be related in some part to the uncertainty over closer economic ties with the European Community during the 1990s, reflected in the initial rejection of the Maastricht Treaty by Danish voters in 1992 and the rejection of the euro in 2000, and in some part by initial uncertainty surrounding the radical labour market reforms starting in 1993. In contrast, the spike in uncertainty in 2005 and 2006 can be associated with the uproar surrounding the political cartoons depicting the Prophet Mohammed that were published in the Danish paper *Jyllands Posten* in 2005, which led to protests throughout the Muslim world and arson attacks on Danish embassies in Syria, Iran and Lebanon.

As far as uncertainty is concerned, our evidence thus suggests that the Great Moderation has overwhelmingly been a global phenomenon. This impression also accords with the general notion that the Great Moderation is attributable to various global causes like improved monetary policy, more sophisticated inventory management techniques, and a dose of good luck in terms of less severe global shocks hitting the world economy (Summers, 2005). The economic crisis of the early 2000s is usually also associated with having been driven by global events like the rising price of oil, the end of the New Economy boom and the 2001 terror attacks. Helbling and Bayoumi (2003) find that whereas the early 1990s have been a distinctive period of relative disconnection between the US and major countries of mainland Europe, the recession of the early 2000s has been highly synchronized among the G-7 countries, with growth cycles all peaking in the year 2000. However, we find little evidence of substantially heightened global uncertainty during this time period. Rather, our results suggest that similar to the 1990s recession, the uncertainty surrounding the early 2000s slowdown has mostly been confined to an increase of national uncertainty in a number of (mostly European) countries, and thus cannot be classified as a global phenomenon. In fact, a number of countries such as Australia, Canada and the UK avoided the recession altogether.

#### **IV. Uncertainty and macroeconomic performance**

As discussed in the introduction, much recent literature is concerned with assessing the impact of macroeconomic uncertainty on economic activity. This section follows up on this discussion by investigating whether and how the global and national measures of uncertainty affect output growth. We are also interested in identifying whether transmission takes place primarily through consumption demand, investment or international trade flows.



To this end, we investigate the dynamic response of these macroeconomic aggregates to global and country-specific uncertainty shocks by estimating VAR models for all  $N = 20$  sample countries on quarterly data from 1970Q2 to 2013Q4.<sup>8</sup> The VAR models we use are unrestricted and contain seven endogenous variables. The variables in estimation order are our uncertainty measure (either global or country-specific), GDP, CPI, consumption, investment, imports and exports.<sup>9</sup> The VAR models contain two lags of each variable as well as an exogenous constant.<sup>10</sup> The error terms are orthogonalized by a Cholesky decomposition.

We estimate all VAR models with either our global uncertainty measure or the country's corresponding country-specific uncertainty measure and analyse the impulse responses of all macrovariables to a one standard deviation shock in uncertainty.

Figure 3 plots the impulse response functions of each country's GDP growth to a global uncertainty shock. All countries display a sharp drop in output growth, ranging from 0.32% for Switzerland to 1.17% for Japan, which are significant for all countries except for Australia, New Zealand and Norway. A subsequent rebound follows between 8 and 15 quarters after the shock in the large majority of our sample countries. This result is in line with Bloom (2009) and Bloom *et al.* (2012), who find a significant drop around 1% in industrial production after a US stock market volatility shock, followed by a recovery and return to the long-run average, with an overshoot of output growth for some quarters.<sup>11</sup>

Figure 4 shows the impulse responses of country-specific uncertainty on output growth. It turns out that compared to the dynamic effects of global uncertainty, the impact of country-specific uncertainty is much less clear-cut. Responses are mostly insignificant, and in the few countries with significant reactions, these disappear very quickly after the shock. Interestingly, the effects of country-specific uncertainty on GDP turn out to be significantly positive at least over some periods of the forecast horizon in Austria and Italy. The possibility of a positive relationship between growth volatility and average growth has first been suggested by Black (1987, Chapters 10 and 13), who argues that the choice of investing in a risky technology may generate higher average growth rates. Blackburn (1999) comes to a similar conclusion based on an endogenous growth model in which business cycle volatility raises the long-run growth rate of the economy. Another argument suggesting a positive relation between output volatility and growth comes from the theory of precautionary savings, where increased risk increases desired savings and thus investment and growth (see Grier *et al.*, 2004, and further references therein).

Figure 5 presents the impulse response functions of consumption to a global uncertainty shock. As expected, the initial responses are all negative, but turn out to be significant in just 7 out of our 20 sample countries. Moreover, only in Australia, Spain, the UK and the US

<sup>8</sup> Because of missing data on investment the VAR models for Australia are estimated for the period from 1974Q4 and UK and Switzerland from 1980Q1 only.

<sup>9</sup> All data are obtained from the OECD Statistics database and are seasonally adjusted. All variables except the uncertainty measure are in annualized log changes. We include the log change in CPI to control for price effects. Results are identical when estimation is performed with real data. Our results are also robust to different variable orderings.

<sup>10</sup> According to the Schwarz criterion  $p = 2$  is the optimal lag length for almost all VARs in our sample.

<sup>11</sup> In the models of Bloom (2009) and Bloom *et al.* (2012) this 'volatility overshoot' arises because of the convexity of hiring and investment decisions.

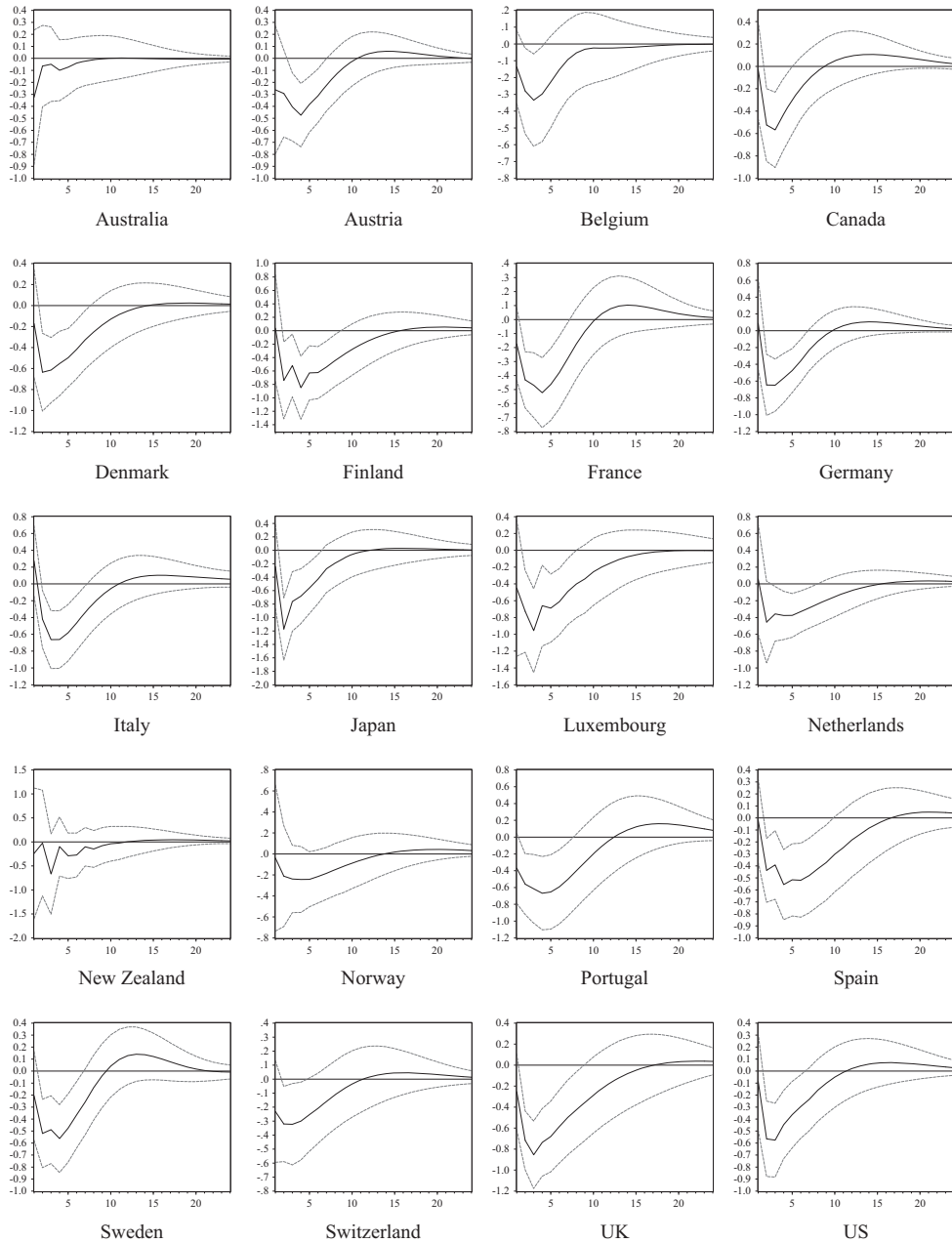


Figure 3. Reaction of GDP to a 1 SD shock in global output growth uncertainty

*Notes:* The dotted lines indicate  $\pm 2$  SE. The horizontal axis denote quarters after the shock. The vertical axis displays the % impact on the dependent variable.

does the impact last for more than one or two quarters. Figure 6 reports the corresponding reactions of consumption to a country-specific uncertainty shock, where about half of all impulse responses are significant. Again some of the responses become significantly positive for some periods of the forecast horizon. Beside Austria and Italy, where similar

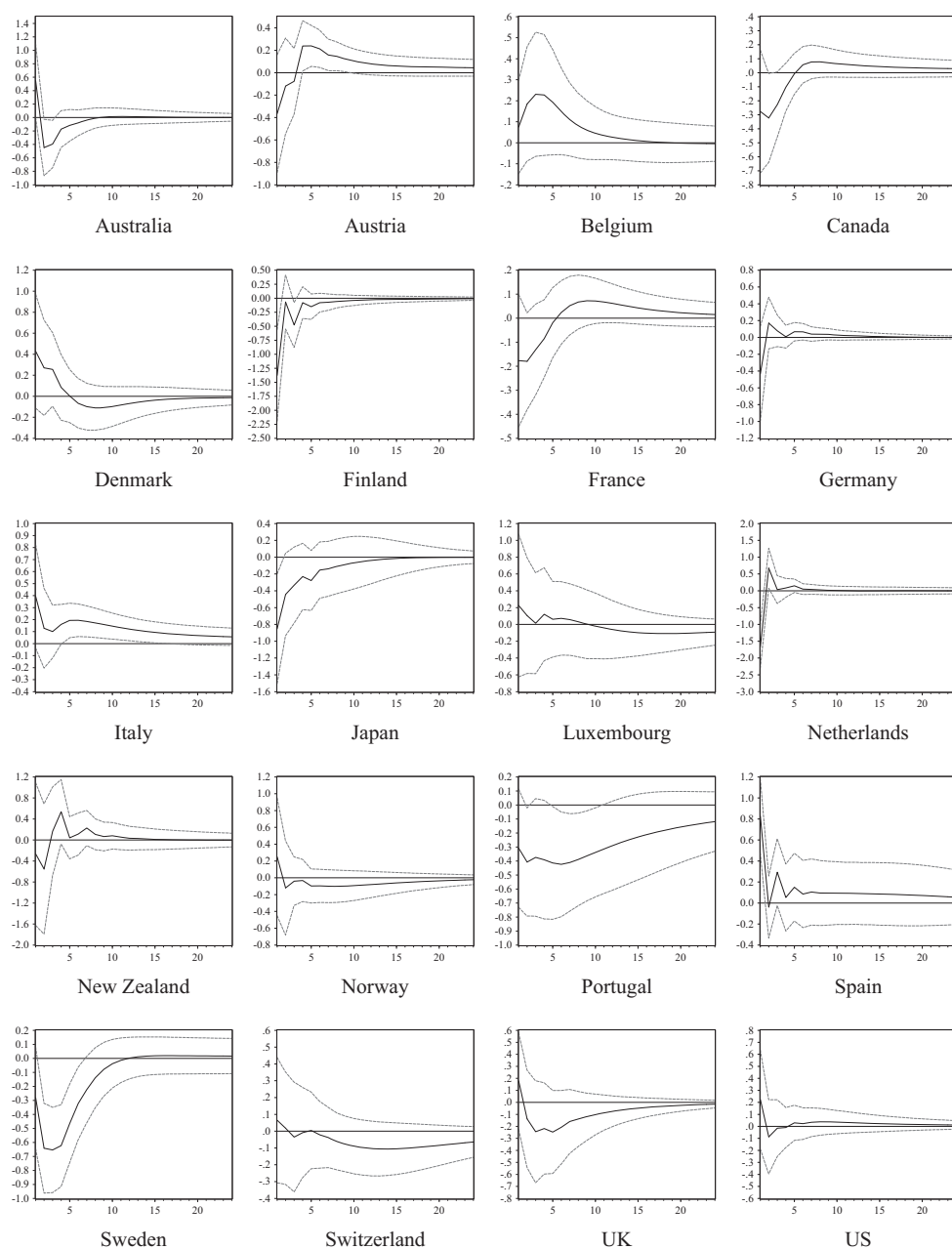


Figure 4. Reaction of GDP to a 1 SD shock in country-specific output growth uncertainty

Notes: The dotted lines indicate  $\pm 2$  SE. The horizontal axis denote quarters after the shock. The vertical axis displays the % impact on the dependent variable.

evidence is obtained for the GDP reactions of Figure 4, consumption rises in response to heightened national uncertainty in Germany for a number of periods, with marginal significance also for Belgium and France. There is some evidence from economic theory for such a positive effect to occur, notably by Mirman (1971), who shows that depending

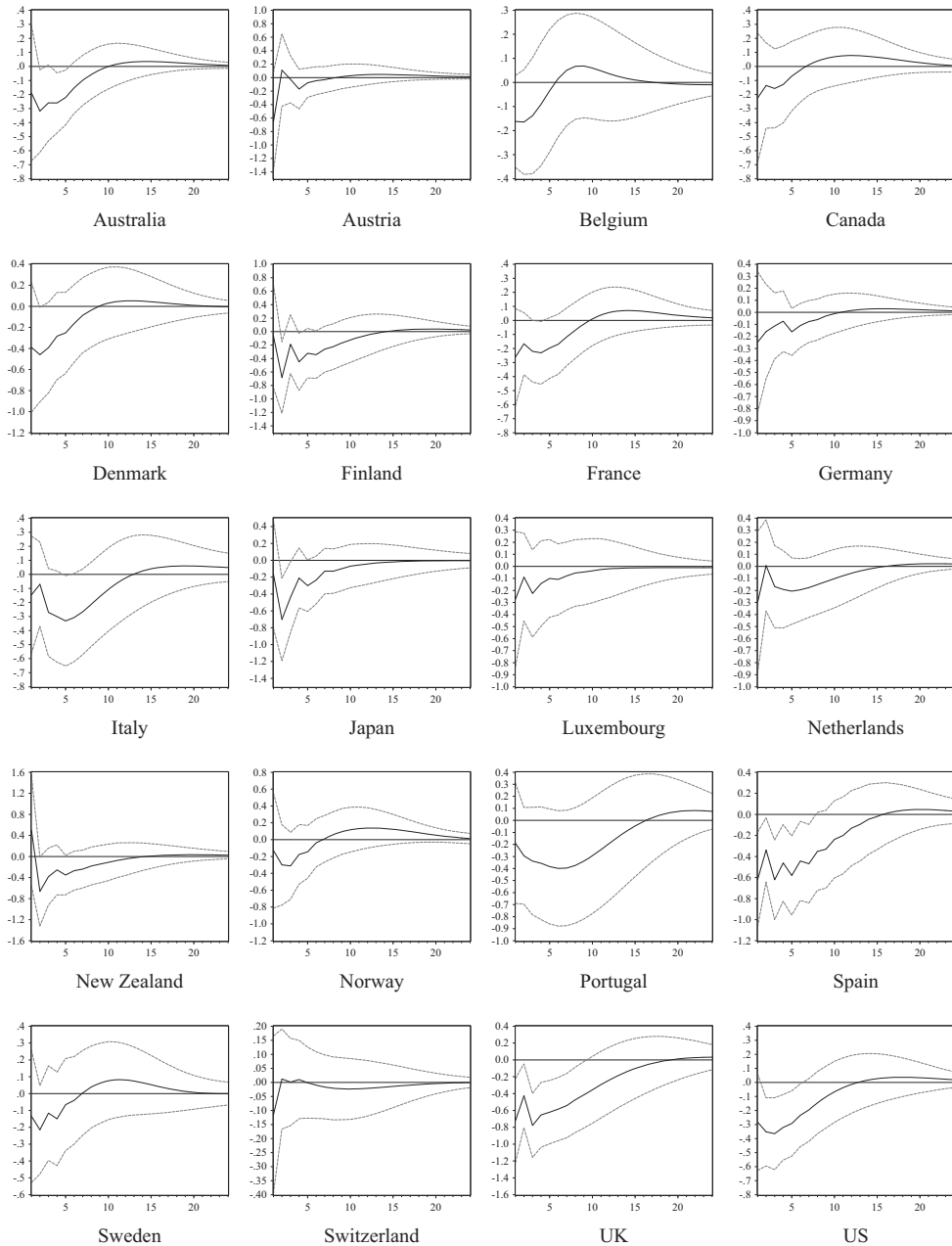


Figure 5. Reaction of consumption to a 1 SD shock in global output growth uncertainty

*Notes:* The dotted lines indicate  $\pm 2$  SE. The horizontal axis denotes the quarters after the shock. The vertical axis displays the % impact on the dependent variable.

on the shape of the utility function, rising uncertainty may generate a desire to consume more initially as a hedge against an uncertain future.

As can be seen from Figure 7, the effect of a global uncertainty shock on investment is unambiguously negative and mostly significant, with the only insignificant results for

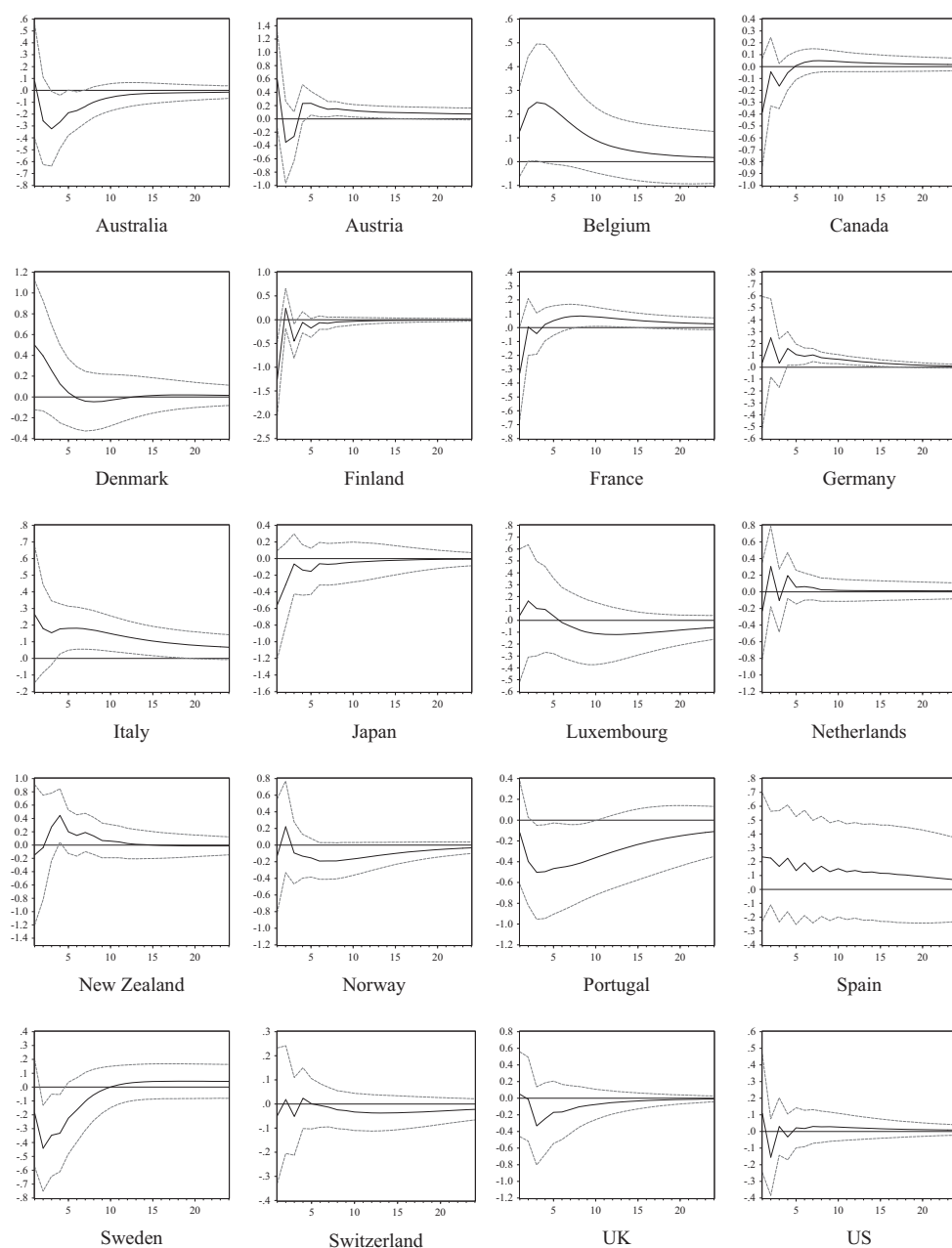


Figure 6. Reaction of consumption to a 1 SD shock in country-specific output growth uncertainty

Notes: The dotted lines indicate  $\pm 2$  SE. The horizontal axis denotes the quarters after the shock. The vertical axis displays the % impact on the dependent variable.

Australia, Finland and Norway. By dropping between 1.04% (Switzerland) and 6.81% (Luxembourg) in peak times, the impact on investment is quantitatively much more important than the output responses reported in Figure 3. This result is in line with previous literature and confirms the investment channel to be an important transmission channel of

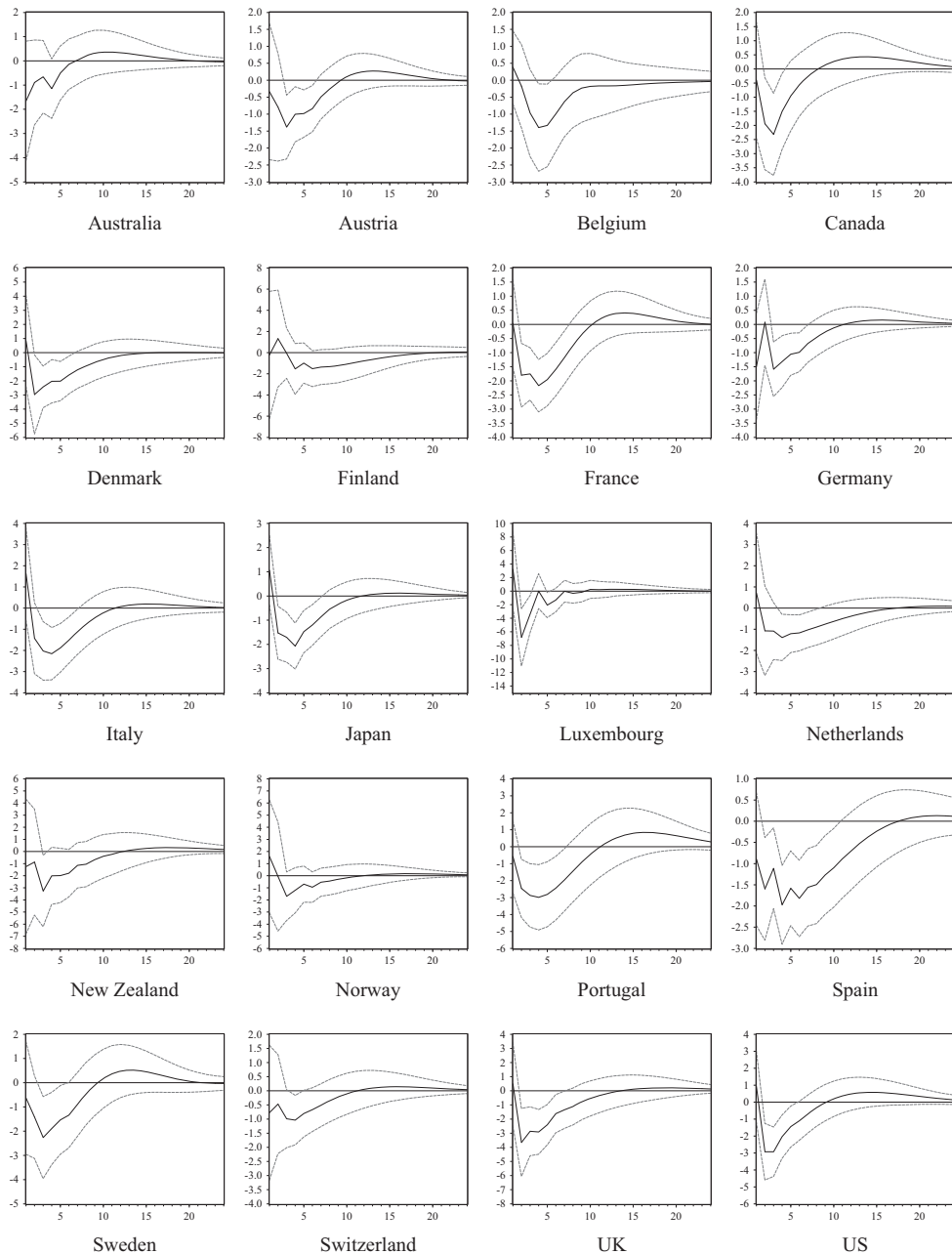


Figure 7. Reaction of investment to a 1 SD shock in global output growth uncertainty

*Notes:* The dotted lines indicate  $\pm 2$  SE. The horizontal axis denotes the quarters after the shock. The vertical axis displays the % impact on the dependent variable.

uncertainty on the real economy. However, as the corresponding responses to a one standard deviation country-specific uncertainty shock displayed in Figure 8 turn out to be mostly insignificant, we find the global rather than country-specific component of uncertainty to be the driving force in the transmission process.

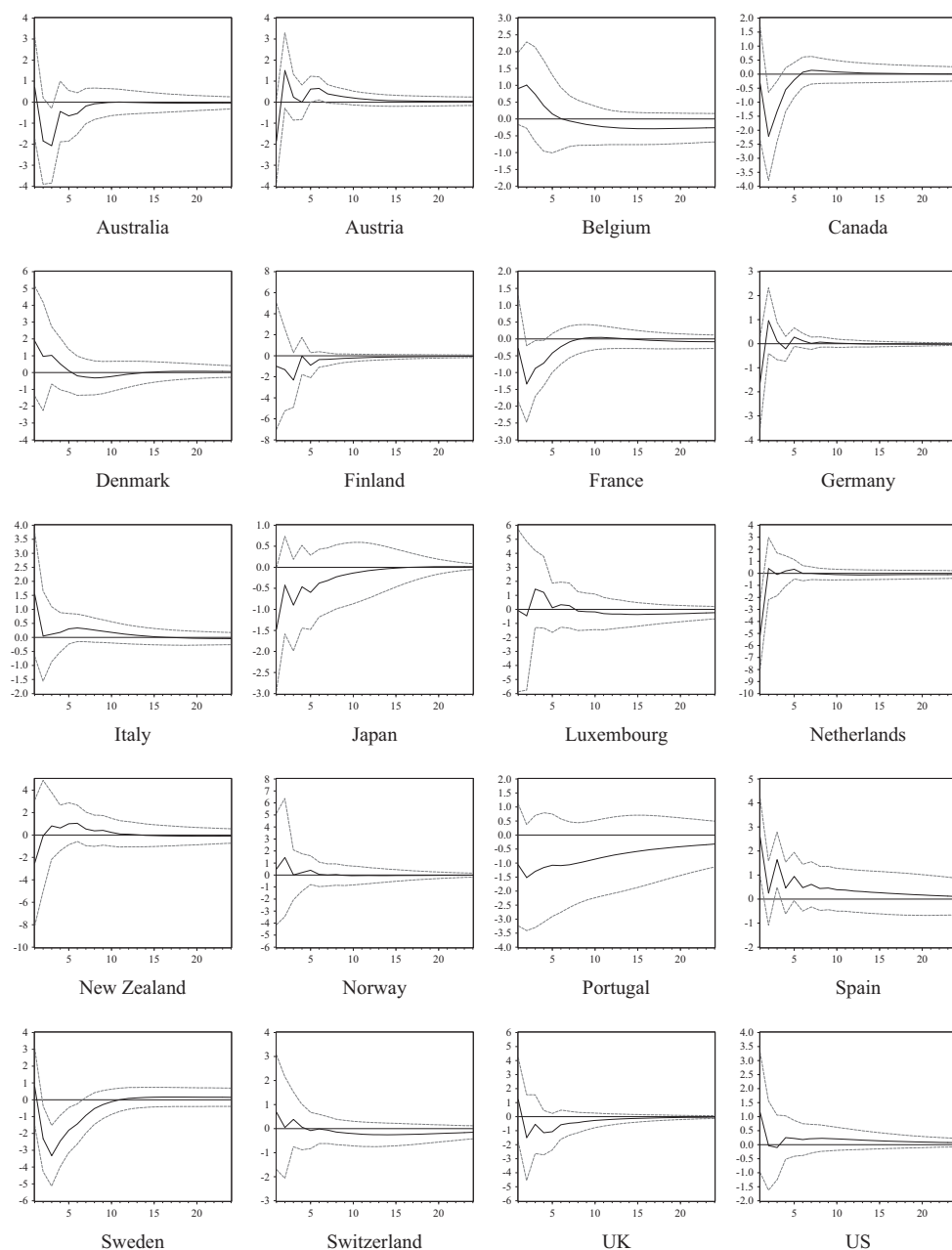


Figure 8. Reaction of investment to a 1 SD shock in country-specific output growth uncertainty  
*Notes:* The dotted lines indicate  $\pm 2$  SE. The horizontal axis denotes the quarters after the shock. The vertical axis displays the % impact on the dependent variable.

Figures 9–12 report on the impact of uncertainty on international trade flows. The response of imports to a global uncertainty shock displayed in Figure 9 is significantly negative in all countries but Norway. Furthermore, it is considerably stronger than the corresponding response of output growth, with imports dropping between 0.97% in Switzer-

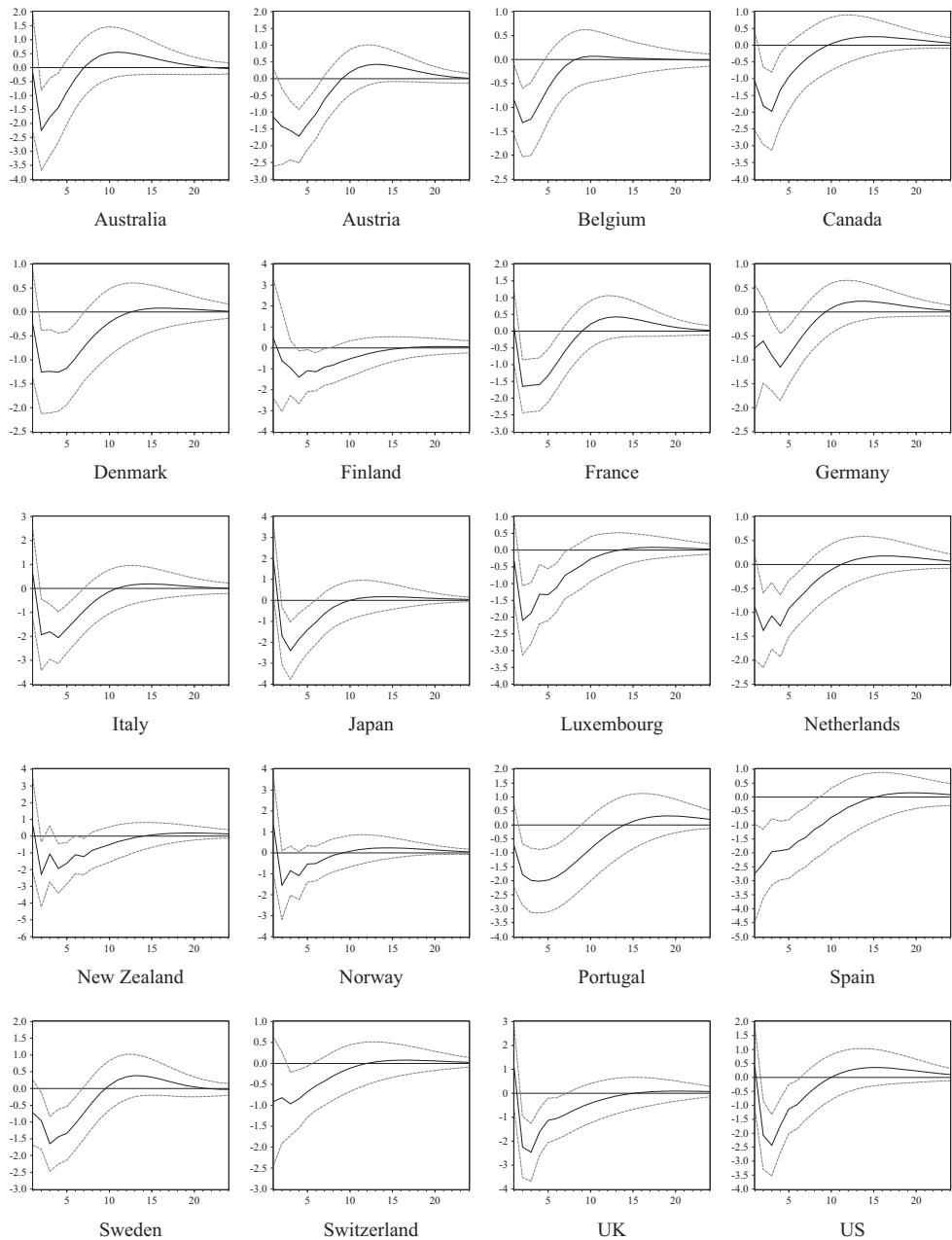


Figure 9. Reaction of imports to a 1 SD shock in global output growth uncertainty

*Notes:* The dotted lines indicate  $\pm 2$  SE. The horizontal axis denote quarters after the shock. The vertical axis displays the % impact on the dependent variable.

land and 2.73% in Spain in its period of peak impact. These results are in line with Novy and Taylor (2014) who estimate the effects of uncertainty shocks on imports to be 5–10 times higher than those on industrial production. The impact of country-specific uncertainty shocks on imports reported in Figure 10 turn out to be again mostly insignificant



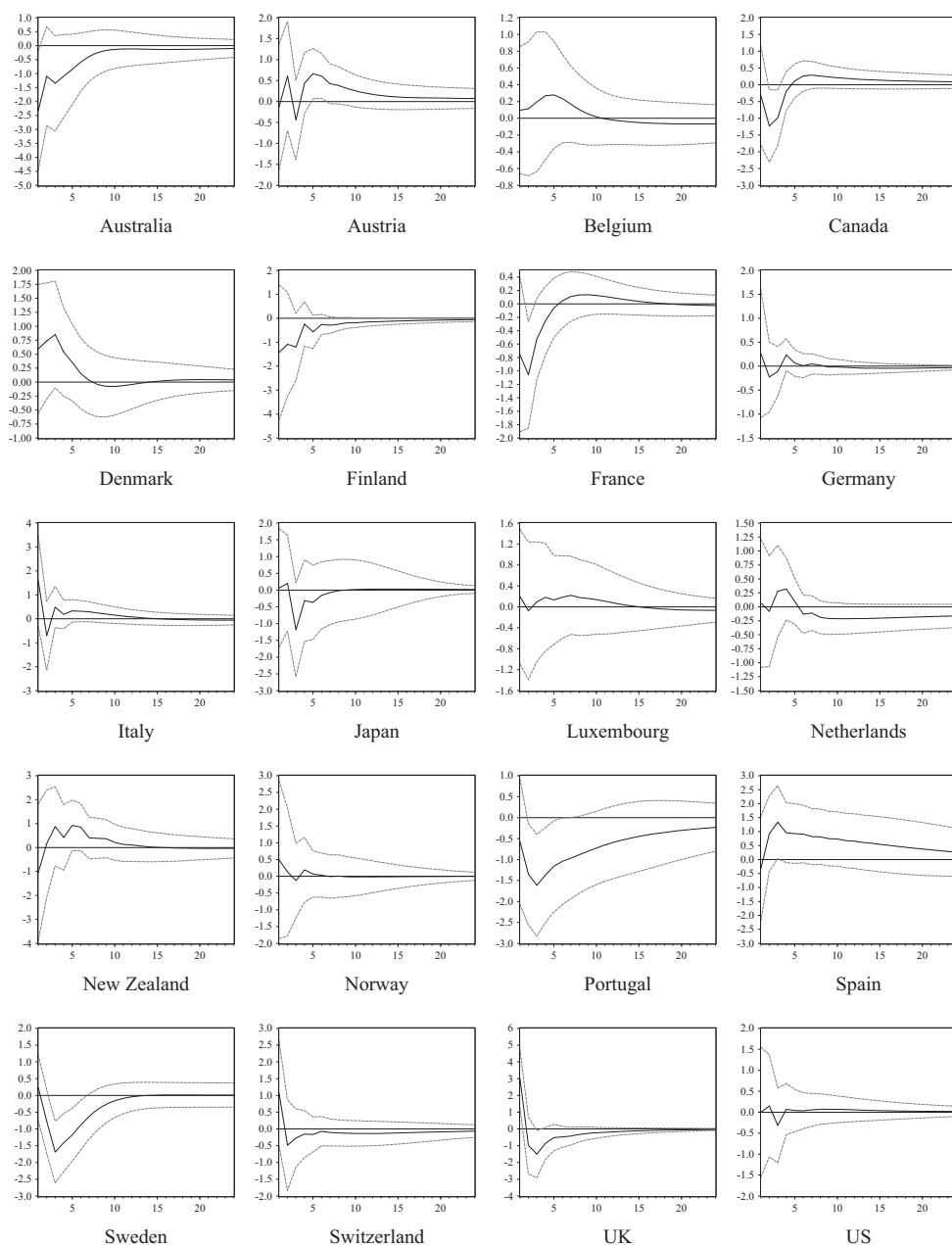


Figure 10. Reaction of imports to a 1 SD shock in country-specific output growth uncertainty

Notes: The dotted lines indicate  $\pm 2$  SE. The horizontal axis denote quarters after the shock. The vertical axis displays the % impact on the dependent variable.

and quantitatively much less important compared to global uncertainty shocks. A similar picture emerges for exports contained in Figures 11 and 12, with quantitatively strong and significantly negative effects of global uncertainty, but much less important and frequently insignificant results for the country-specific components of uncertainty.

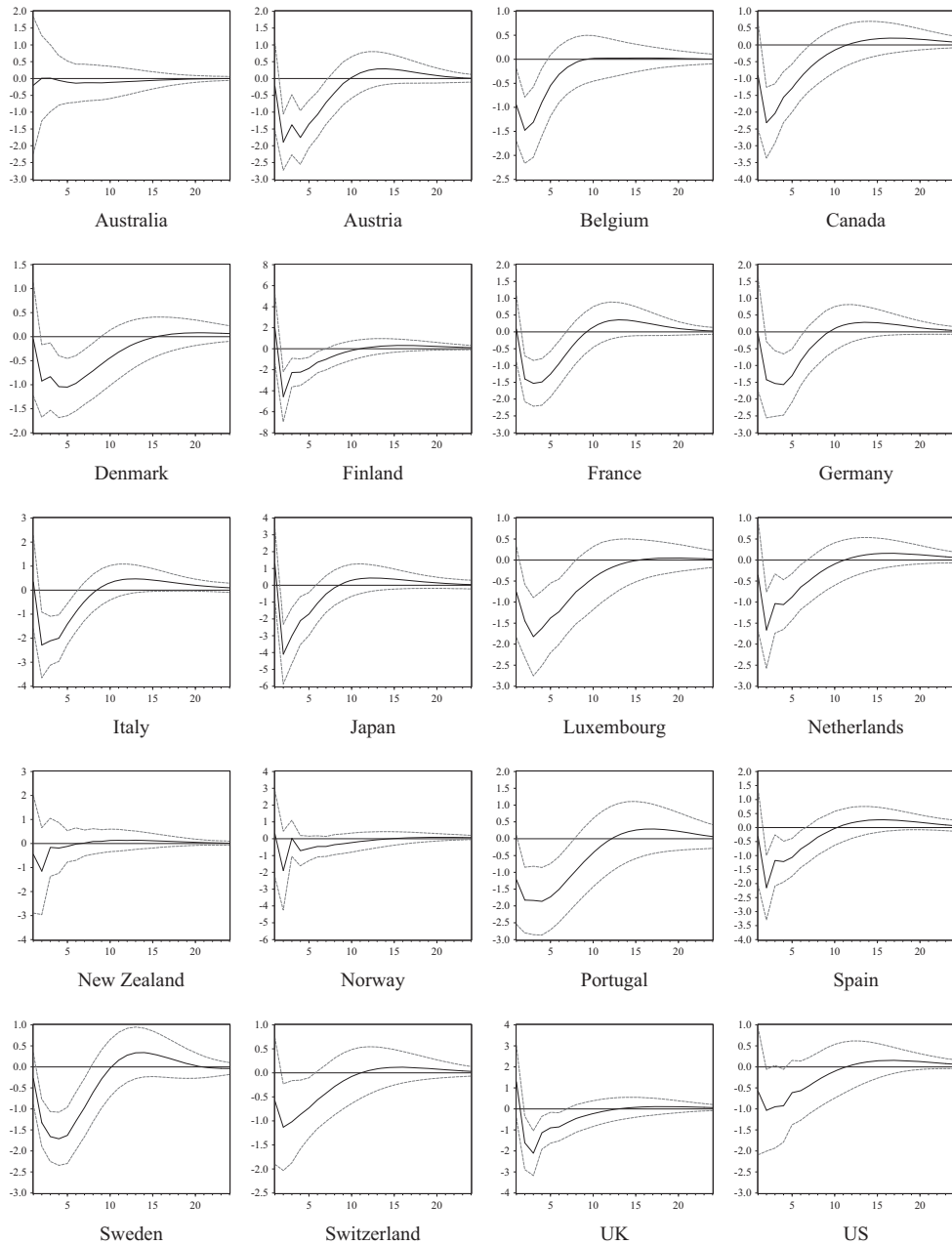


Figure 11. Reaction of exports to a 1 SD shock in global output growth uncertainty

Notes: The dotted lines indicate  $\pm 2$  SE. The horizontal axis denote quarters after the shock. The vertical axis displays the % impact on the dependent variable.

## V. Conclusion

A growing body of theoretic and empirical literature points to a negative relationship between uncertainty and economic performance. However, so far this literature mostly

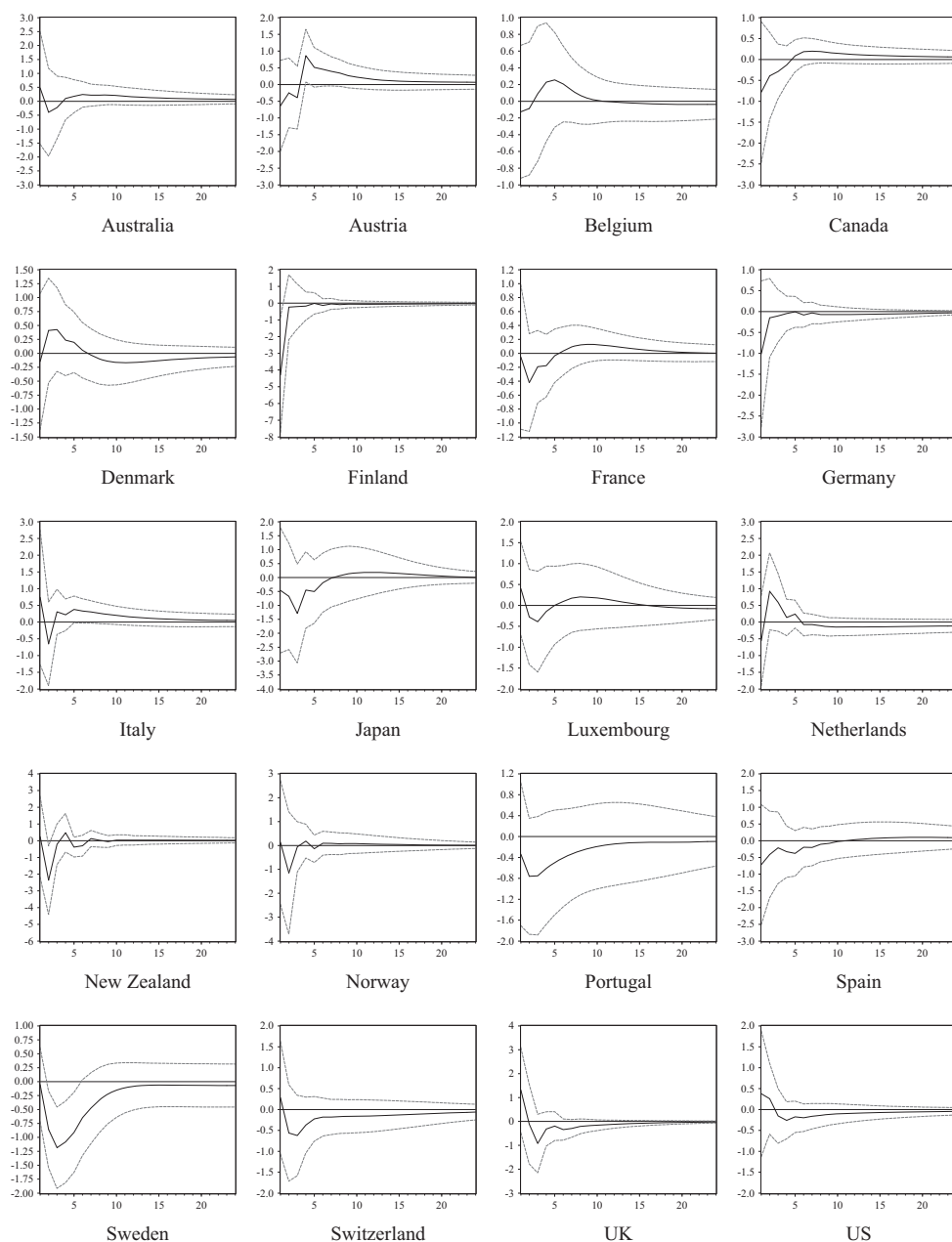


Figure 12. Reaction of exports to a 1 SD shock in country-specific output growth uncertainty  
*Notes:* The dotted lines indicate  $\pm 2$  SE. The horizontal axis denote quarters after the shock. The vertical axis displays the % impact on the dependent variable.

focuses on the national rather than the global dimension of uncertainty. In this paper, we identify global and country-specific measures of output growth uncertainty for a large OECD country sample by means of a DFM with stochastic volatility. Our results show evidence for major bouts of global uncertainty in the early 1970s and late 2000s, and a

number of periods with elevated levels of either global or national uncertainty, particularly in the early 1980s, 1990s and 2000s. Furthermore, we find evidence suggesting that the Great Moderation has overwhelmingly been a global phenomenon.

We then assess the impacts of global uncertainty vs. country-specific uncertainty on macroeconomic performance. The literature proposes different transmission channels through which uncertainty shocks might have an influence on real macroeconomic variables. By separately analyzing consumption, investment and the trade components of aggregate demand we shed light on the particular channels through which uncertainty affects the real economy. To this end, we estimate VAR models to analyse the response of output, consumption, investment, imports and exports to global and domestic uncertainty shocks. The VAR impulse responses reveal that global uncertainty is the major driver of macroeconomic performance in most countries, whereas the impact of national uncertainty is small and frequently insignificant. We also find that uncertainty is transmitted primarily through investment and trade flows rather than through consumption demand.

## Appendix A. Details on the estimation of stochastic volatilities

In this appendix, we give a more detailed explanation of the filtering and sampling mechanism to obtain the stochastic volatilities in step 3 and 4 of the Gibbs sampler routine. To estimate the stochastic volatility parameters, we follow an approach developed by Kim, Shephard and Chib (1998) and refined by Omori *et al.* (2007) that provides a complete MCMC simulation-based analysis on the estimation of stochastic volatility models.

As can be seen in the Gibbs sampler description in the main part of this paper, we filter and sample the volatility series for both the common factor  $h_t$  and all country-specific factors  $g_{it} = \{g_{1t}, \dots, g_{Nt}\}$  in two separate steps. This is possible due to the fact that all stochastic volatility components in our model are independent of each other. A key feature of the stochastic volatility components  $\exp(h_t)\xi_t$  and  $\exp(g_{it})\sigma_{\eta_i}\eta_{it}$  is that they are nonlinear but can be transformed into linear components by taking the logarithm of their squares

$$\log(\exp(h_t)\xi_t)^2 = 2h_t + \log \xi_t^2, \quad (\text{A.1a})$$

$$\log(\exp(g_{it})\sigma_{\eta_i}\eta_{it})^2 = 2g_{it} + \log \sigma_{\eta_i}^2 + \log \eta_{it}^2, \quad (\text{A.1b})$$

where the error terms  $\log \xi_t^2$  and  $\log \eta_{it}^2$  follow a log-chi-square density with expected value  $-1.2704$  and variance  $4.93$ . Following Omori *et al.* (2007), we approximate the linear models in (A.1) by an offset mixture time series model as

$$h_t^* = 2h_t + \xi_t^*, \quad (\text{A.2a})$$

$$g_{it}^* = 2g_{it} + \log \sigma_{\eta_i}^2 + \eta_{it}^*, \quad (\text{A.2b})$$

where  $h_t^* = \log((\exp(h_t)\xi_t)^2 + c)$ , and  $g_{it}^* = \log((\exp(g_{it})\sigma_{\eta_i}\eta_{it})^2 + c)$ . The offset constant  $c$  is set  $c = .001$ , and the distributions of  $\xi_t^*$  and  $\eta_{it}^*$  being a mixture of  $M$  normal densities.

The mixture approximation has the form

$$f(\zeta_t^*) = \sum_{j=1}^M q_j f_N(\zeta_t^* | m_j - 1.2704, v_j^2), \quad (\text{A.3a})$$

$$f(\eta_{it}^*) = \sum_{j=1}^M q_j f_N(\eta_{it}^* | m_j - 1.2704, v_j^2), \quad (\text{A.3b})$$

where  $f(\zeta_t^*)$  and  $f(\eta_{it}^*)$  denote normally distributed density functions with mean  $m_j - 1.2704$  and variance  $v_j^2$ . Equivalently, the mixture densities can also be written in terms of the component indicator variables  $s_t$  and  $k_{it}$  as

$$\zeta_t^* | (s_t = j) \sim \mathcal{N}(m_j - 1.2704, v_j^2), \quad \text{with } \Pr(s_t = j) = q_j, \quad (\text{A.4a})$$

$$\eta_{it}^* | (k_{it} = j) \sim \mathcal{N}(m_j - 1.2704, v_j^2), \quad \text{with } \Pr(k_{it} = j) = q_j. \quad (\text{A.4b})$$

We follow Omori *et al.* (2007) by selecting  $M = 10$  and using the parameters  $\{q_j, m_j, v_j^2\}$  in their Table 1 to make the approximation of the mixture distributions to the log-chi-square distribution sufficiently good. The conditional probability mass functions for  $s_t$  and  $k_{it}$  are given by

$$\Pr(s_t = j | h_t, \zeta_t^*) \propto q_j f_N(\zeta_t^* | 2h_t + \log \sigma_\xi^2 + m_j - 1.2704, v_j^2), \quad (\text{A.5a})$$

$$\Pr(k_{it} = j | g_{it}, \eta_{it}^*) \propto q_j f_N(\eta_{it}^* | 2g_{it} + \log \sigma_{\eta_i}^2 + m_j - 1.2704, v_j^2). \quad (\text{A.5b})$$

Below we use the notation  $s = \{s_t\}_{t=1}^T$  and  $k_i = \{k_{it}\}_{t=1}^T$ . Following Del Negro and Primiceri (2013), in this block we first sample the mixture indicators  $s_t$  and  $k_{it}$  from their conditional probability mass functions (A.5a) and (A.5b).

Next, we filter and sample the stochastic volatilities  $h_t$  and  $g_{it}$ , conditioning on the transformed states  $h_t^* = \log(\zeta_t^2 + 0.001)$  and  $g_{it}^* = \log(\eta_{it}^2 + 0.001)$ , on the mixture indicators  $s_t$  and  $k_{it}$  and on the relevant hyperparameters. The state space representation of the model for  $h_t$  is given by:

$$\overbrace{[h_t^* - (m_{s_t} - 1.2704)]}^{y_t^*} = \overbrace{[2]}^Z \overbrace{[h_t]}^{\alpha_t} + \overbrace{[1]}^{s_t} \overbrace{[\zeta_t^*]}^{\varepsilon_t}, \quad (\text{A.6})$$

$$\overbrace{[h_{t+1}]}^{\alpha_{t+1}} = \overbrace{[1]}^T \overbrace{[h_t]}^{\alpha_t} + \overbrace{[\sigma_\gamma]}^{K_t} \overbrace{[\gamma_t]}^{\psi_t}, \quad (\text{A.7})$$

and  $H_t = [v_{s_t}^2]$ . For given values of  $s_t$ , sampled in block 3,  $m_{s_t}$  and  $v_{s_t}^2$  are taken from Table 1 in Omori *et al.* (2007) and  $\tilde{\zeta}_t^* = \zeta_t^* - (m_{s_t} - 1.2704)$  is  $\zeta_t^*$  recentred around zero. After drawing  $h_t$  we normalize  $\frac{1}{T} \sum_{t=1}^T \exp(h_t) = 1$ .

Similarly, the state space representation of the model for  $g_{it}$  is given by the following:

$$\overbrace{[g_{it}^* - (\log \sigma_{\eta_i}^2 + m_{k_{it}} - 1.2704)]}^{y_{it}^*} = \overbrace{[2]}^Z \overbrace{[g_{it}]}^{\alpha_{it}} + \overbrace{[1]}^{s_t} \overbrace{[\eta_{it}^*]}^{\varepsilon_{it}}, \quad (\text{A.8})$$

$$\underbrace{[g_{i,t+1}]}_{\alpha_{t+1}} = \underbrace{[1]}_T \underbrace{[g_{it}]}_{\alpha_t} + \underbrace{[\sigma_{\omega_i}]}_{K_t} \underbrace{[\omega_{it}]}_{\psi_t}, \quad (\text{A.9})$$

and  $H_t = [v_{w_{it}}^2]$ . For given values of  $k_{it}$ , sampled in block 4,  $m_{k_{it}}$  and  $v_{k_{it}}^2$  are taken from Table 1 in Omori *et al.* (2007) and  $\tilde{\eta}_{it}^* = \eta_{it}^* - (m_{k_{it}} - 1.2704)$  is  $\eta_{it}^*$  recentred around zero. After drawing  $g_{it}$  we normalize  $\frac{1}{T} \sum_{t=1}^T \exp(g_{it}) = 1$ .

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